

Analysis of NBS: Ecological & economic outcomes

CUNY EES 79903

Seminar on Nature-Based Solutions

Brett Branco and Kyle McKay

September 29, 2026

Lecture Overview:

- 415-430 – Housekeeping: field trip, projects
- 430-550 – Extras: conference report-out, NBS in Battery Park
- 500-615 – Discussion: Feagin et al. (2022), Stanford et al. (2025), Reidenbach et al. (2026)

	Leader #1	Leader #2
Sep 8: NBS fundamentals	Grace	Jesse
Sep 15: NBS fundamentals	Michelle	Shahela
Sep 22: Structured decision making and community engagement	Jesse	Aurelia
Sep 29: Analysis – ecological	Carol	Julia
Oct 6: Analysis – co-benefits	Shahela	Aurelia
Oct 20: Analysis – engr performance	Grace	Michelle
Oct 27: Analysis – engr performance	Abdellah	Julia
Nov 3: Alternatives analysis	Aurelia	Abdellah
Nov 10: Performance-based design	Aaryan	Michelle
Nov 17: Constructability	Grace	Aaryan
Nov 24: Performance monitoring	Shahela	Jesse
Dec 1: Performance monitoring	Julia	Carol
Dec 8: Life cycle analysis	Aaryan	Abdellah
Dec 15: Course synthesis	Kyle	Brett

Field trip planning

October 11th – 900am-300pm

- 845 – Meet at Brooklyn College
- 900 – VAN LEAVES
- 945-1045 – Rockaways – Beach 43rd St. / USACE project
- 1130-100 – Broad Channel – Lunch, West Pond, marsh islands, Sunset Cove marsh, thin layer placement sites
- 100-230 – Shirley Chisolm State Park
- 300 – Adjourn from Brooklyn College

Project Sign-ups:

Everyone gets their first pick!

Topic	First option	Second option
Qualitative foundations of project planning	Jesse	Grace
Facilitation, engagement, and teaming	Aurelia	Jesse
Ecological analysis of NBS	Julia	
Economic analysis of NBS		
Co-benefits analysis of NBS		
Engineering analysis of NBS	Abdellah	
Alternatives analysis and structured decision making		
Performance-based design	Aaryan	
Constructability	Michelle	
Performance monitoring		
Project establishment		
Asset management	Grace	

Project Timeline

- Oct 20 – Recommended timeframe to compile initial references
 - McKay and Branco are happy to help seed your literature reviews 😊
- Nov 3 – Recommended timeframe for outlining ideas
- Nov 17 – Initial submission of traceable account
- Nov 24 – Review of traceable account by classmates
 - Each person will review two other accounts and provide written feedback
- Dec 8 – Final submission of traceable account
- Dec 15 – Course synthesis (McKay)

Conference Report-out: Restore America's Estuaries

Things I heard about estuaries
and nature-based solutions:

- Relationships facilitate actions as well as fundraising, permitting, idea sharing,...
- Uncertainties in the federal role, but progress continues
- SF and NYC have interesting parallels in approach

Conference strategies:

- Review the agenda
- Set up meetings and meals
- Cruise the posters and booths
- Be courageous
- Make time to fill your battery
- FOLLOW-UP

Nature-based Solution?



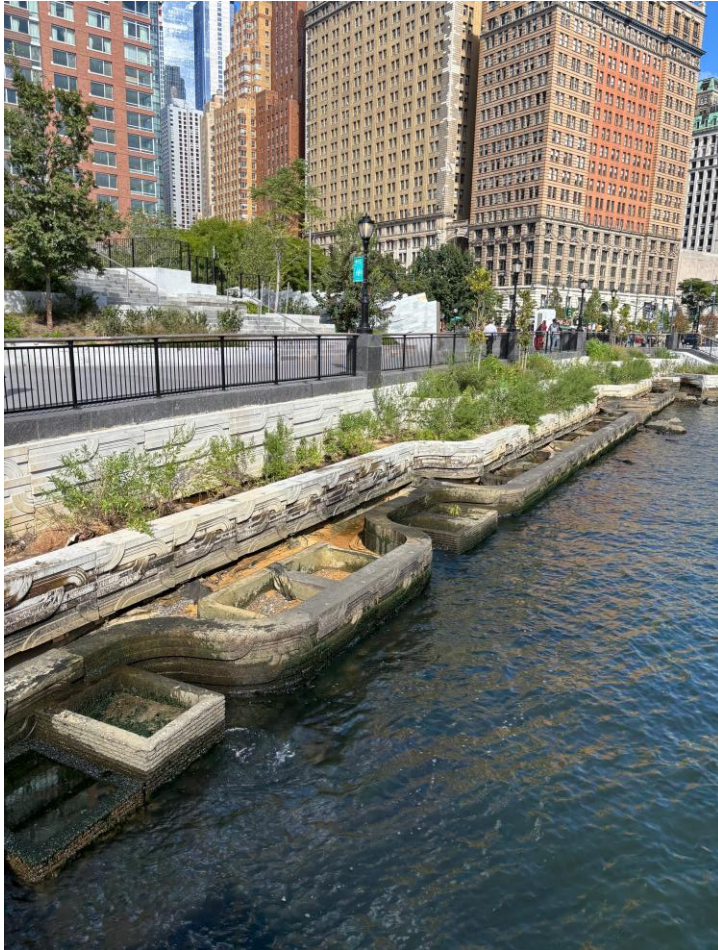
<https://econcretetech.com/projects/south-battery-park-city-resiliency-project/>

Creating a Flourishing Habitat Amidst Steel and Stone

<https://waterfrontalliance.org/2024/01/23/south-battery-park-city-resiliency-sbpcr-achieves-13th-wedg-verification-in-the-nation/>

At the south end of the site along the Pier A Inlet, new habitat space will emerge. Today, stone riprap provides an unwelcoming and unproductive shoreline edge. Once built, this section of shoreline will consist of plantings on stepped terraces in the intertidal zone that will bring marine life to the site. An overwater platform will use grating to allow viewing of the habitat space while also letting light penetrate the water which is key to the success of near-shore in-water habitats. The ecological improvements will provide habitat for striped bass, scup, blue fish, oysters, lobsters, blue crabs, ring-billed gulls, and great blue herons. The lowest step in the terrace integrates tide pools and intertidal grasses that will at times be below tide and at times above. Low and high marsh plantings make up the upper terraces.

The Reality



Thought Questions

1. Was it a nature-based solution?
2. Regardless of answer to 1, why did it fail?
3. Regardless of answer to 1, how do outcomes like this potentially affect the ability to do more nature-based interventions along the shoreline in NYC?
4. Are projects like this too small in scale to have any co-benefits even if they perform as designed?

Last Week

Ibrahim, A., Marshall, K., Carmen, E., Blackstock, K.L. and Waylen, K.A., 2025. Raising standards for stakeholder engagement in Nature-based Solutions: Navigating the why, when, who and how. *Environmental Science & Policy*, 163, p.103971.

Crawford, B.A., Maerz, J.C. and Moore, C.T., 2026. Navigating multiple objectives, partners, and jurisdictions: Decision analysis for range-wide conservation planning for an at-risk species. *Conservation Science and Practice*, p.e70291.

Pathak A., Del Angel D.C., and Hale C. 2026. Co-developing nature-based solutions for coastal resilience: A case study from Refugio County, Texas. *Estuaries and Coasts*, 49, 31.

Contrasting views of engagement and decision-making

The papers:

- Ibrahim et al. (2025): Principles centered on who, what, when, where, and why. Engagement as a means to other NBS goals.
- Pathak et al. (2026): Richness in intimate, local interactions. Facilitating input in early-stage conceptualization
- Crawford et al. (2026): Meaty model-centric approach for transparent and repeatable decisions. Technical stakeholder audience to guide decisions.

Themes?

- Tailoring methods to project needs
- Different parts of the NBS process require voice for different audiences
- Principles sounds great on paper, but translating this methods into reality might be as much art as science.

Paper Discussion

Feagin, R.A., Bridges, T.S., Bledsoe, B., Losos, E., Ferreira, S., Corwin, E., Lodder, Q., Beck, M.W., Reguero, B., Sutton-Grier, A. and Figlus, J., 2021. Infrastructure investment must incorporate Nature's lessons in a rapidly changing world. *One Earth*, 4(10), pp.1361-1364.

Stanford, G.O., Ferreira, S., Landry, C.E. and Blachly, B., 2025. Economic valuation of ecosystem services and natural infrastructure: A quantitative review of the literature. *Ecosystem Services*, 75, p.101754.

Reidenbach, M.A., Li, M., Rose, K.A., Tomiczek, T., Morris, J., Palinkas, C.M., Staver, L.W., Nardin, W., Gray, M.W., Lee, S.B. and Sutton-Grier, A.E., 2025. Performance Evaluation of Natural and Nature-Based Features for Coastal Protection and Co-Benefits. *Annual Review of Marine Science*, 18.

Feagin et al. (2021)

- What motivated this paper?
- How do other political priorities create an opportunity cost problem for NBS?
- What are the philosophical and numerical implications of design life and discount rates?
- What co-benefits should “count” for a given agency?

Stanford et al. (2025)

- Are these trends in the fields of practice surprising?
- Does a literature review miss many forms of economic analysis?
- What's an example of marine natural infrastructure?

Reidenbach et al. (2026)

- What do you think of the definition of performance? What is the difference between engineering performance and other forms?
- The authors differential primary and secondary project outcomes. How does this align with your mental model of NBS?
- What does a robust co-benefit assessment framework look like?

General Discussion

- Different types of papers: bibliometric review, synthesis, commentary
- What are co-benefits?
- Why do we need to assess projects at all?

Next Week

González-García, A., Palomo, I., Codemo, A., Rodeghiero, M., Dubo, T., Vallet, A. and Lavorel, S., 2025. Co-benefits of nature-based solutions exceed the costs of implementation. *Cell Reports Sustainability*, 2(3).

Sadaf, A. and Bruck, J., 2026. Optimizing shoreline suitability models framework: integrating advanced GIS methodologies at Aberdeen Proving Ground, Maryland. *Frontiers in Environmental Science*, 14, p.1812979.

Chrysoulakis, N., Somarakis, G., Stagakis, S., Mitraka, Z., Wong, M.S. and Ho, H.C., 2021. Monitoring and evaluating nature-based solutions implementation in urban areas by means of earth observation. *Remote Sensing*, 13(8), p.1503.